

JEE Mains 2019 Chapter wise Question Bank

Fluid Mechanics - Questions

Q1

The top of a water tank is open to air and its water level is maintained. It is giving out 0.74m^3 water per minute through a circular opening of 2 cm radius in its wall. The depth of the centre of the opening from the level of water in the tank is close to:

- (1) 6.0 m (2) 4.8 m
(3) 9.6 m (4) 2.9 m

9 Jan Evening

Q2

Water flows into a large tank with flat bottom at the rate of $10^{-4}\text{ m}^3\text{ s}^{-1}$. Water is also leaking out of a hole of area 1 cm^2 at its bottom. If the height of the water in the tank remains steady, then this height is:

- (1) 5.1 cm (2) 7 cm
(3) 4 cm (4) 9 cm

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Q3

A long cylindrical vessel is half filled with a liquid. When the vessel is rotated about its own vertical axis, the liquid rises up near the wall. If the radius of vessel is 5 cm and its rotational speed is 2 rotations per second, then the difference in the heights between the centre and the sides, in cm, will be:

- (1) 2.0 (2) 0.1
(3) 0.4 (4) 1.2

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Q4

Water from a pipe is coming at a rate of 100 liters per minute. If the radius of the pipe is 5 cm, the Reynolds number for the flow is of the order of: (density of water = 1000 kg/m^3 , coefficient of viscosity of water = 1 mPa s)

- (1) 10^3 (2) 10^4 (3) 10^2 (4) 10^6

8 April Morning

Q5

A wooden block floating in a bucket of water has $\frac{4}{5}$ of its volume submerged. When certain amount of an oil poured into the bucket, it is found that the block is just under the oil surface with half of its volume under water and half in oil. The density of oil relative to that of water is:

- (1) 0.5 (2) 0.8 (3) 0.6 (4) 0.7

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Q6

A cubical block of side 0.5 m floats on water with 30% of its volume under water. What is the maximum weight that can be put on the block without fully submerging it under water? [Take, density of water = 10^3 kg/m^3]

- (1) 46.3 kg (2) 87.5 kg (3) 65.4 kg (4) 30.1 kg

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Q7

A submarine experiences a pressure of $5.05 \times 10^6\text{ Pa}$ at depth of d_1 in a sea. When it goes further to a depth of d_2 , it experiences a pressure of $8.08 \times 10^6\text{ Pa}$. Then $d_2 - d_1$ is approximately (density of water = 10^3 kg/m^3 and acceleration due to gravity = 10 ms^{-2}):

- (1) 300 m (2) 400 m (3) 600 m (4) 500 m

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Q8

Water from a tap emerges vertically downwards with an initial speed of 1.0 ms^{-1} . The cross-sectional area of the tap is 10^{-4} m^2 . Assume that the pressure is constant throughout the stream of water and that the flow is streamlined. The cross-sectional area of the stream, 0.15 m below the tap would be : [Take $g = 10 \text{ ms}^{-2}$]

- (1) $2 \times 10^{-5} \text{ m}^2$ (2) $5 \times 10^{-5} \text{ m}^2$
(3) $5 \times 10^{-4} \text{ m}^2$ (4) $1 \times 10^{-5} \text{ m}^2$

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Q9

A solid sphere, of radius R acquires a terminal velocity v_1 when falling (due to gravity) through a viscous fluid having a coefficient of viscosity η . The sphere is broken into 27 identical solid spheres. If each of these spheres acquires a terminal velocity, v_2 , when falling through the same fluid, the ratio (v_1/v_2) equals:

- (1) 9 (2) $1/27$ (3) $1/9$ (4) 27

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Fluid Mechanics - Answers

Q1

(2) Here, volume flow rate

$$= \frac{0.74}{60} = \pi r^2 v = (\pi \times 4 \times 10^{-4}) \times \sqrt{2gh}$$

$$\Rightarrow \sqrt{2gh} = \frac{74 \times 100}{240\pi} \Rightarrow \sqrt{2gh} = \frac{740}{24\pi}$$

$$\Rightarrow 2gh = \frac{740 \times 740}{24 \times 24 \times 10} (\because \pi^2 = 10)$$

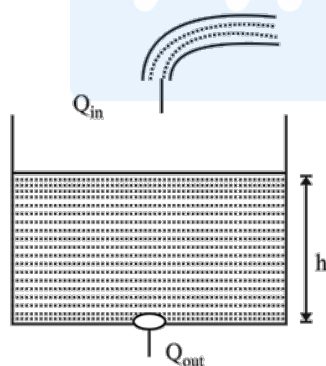
$$\Rightarrow h = \frac{74 \times 74}{2 \times 24 \times 24} \approx 4.8 \text{ m}$$

i.e., The depth of the centre of the opening from the level of water in the tank is close to 4.8 m

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Q2

(1)



Since height of water column is constant therefore, water inflow rate (Q_{in}) = water outflow rate

$$Q_{in} = 10^{-4} \text{ m}^3 \text{ s}^{-1}$$

$$Q_{out} = Au = 10^{-4} \times \sqrt{2gh}$$

$$\therefore 10^{-4} = 10^{-4} \times \sqrt{20 \times h}$$

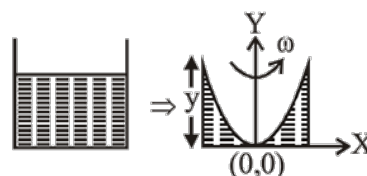
$$\therefore h = \frac{1}{20} \text{ m} = 5 \text{ cm}$$

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Q3

(1) Using $v^2 = u^2 + 2gy$ [$\because u = 0$ at $(0,0)$]

$$v^2 = 2gy \quad [\because v = \omega x]$$



$$\Rightarrow y = \frac{\omega^2 x^2}{2g} = \frac{(2 \times 2\pi)^2 \times (0.05)^2}{20} \approx 2 \text{ cm}$$

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Q4

(2) Rate of flow of water (V) = 100 lit/min

$$= \frac{100 \times 10^{-3}}{60} \times \frac{5}{3} \times 10^{-3} \text{ m}^3$$

 $\therefore$ Velocity of flow of water (v)

$$= \frac{V}{A} = \frac{5 \times 10^{-3}}{3 \times (5 \times 10^{-2})^2}$$

$$= \frac{10}{15\pi} = \frac{2}{3\pi} \text{ m/s} = 0.2 \text{ m/s}$$

$$\therefore \text{Reynold number } (N_R) = \frac{Dvp}{\eta}$$

$$= \frac{(10 \times 10^{-2}) \times \frac{2}{3\pi} \times 1000}{1} = 2 \times 10^4$$

$$\text{Order of } N_R = 10^4$$

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Q5

Fluid Mechanics

$$(3) \quad Mg = \left(\frac{4V}{5}\right) \rho \omega g$$

$$\text{or } \left(\frac{M}{V}\right) = \frac{4\rho\omega}{5}$$

$$\text{or } \rho = \frac{4\rho\omega}{5}$$

When block floats fully in water and oil, then

$$Mg = F_{b1} + F_{b2}$$

$$(\rho V)g = \left(\frac{V}{2}\right) \rho_{\text{oil}} g + \frac{V}{2} \rho \omega g$$

$$\text{or } \rho_{\text{oil}} = \frac{3}{5} \rho \omega = 0.6\rho\omega$$

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Q6

(2) When a body floats then the weight of the body = upthrust

$$\therefore (50)^3 \times \frac{30}{100} \times (1) \times g = M_{\text{cube}} g \quad \dots(i)$$

Let m mass should be placed, then

$$(50)^3 \times (1) \times g = (M_{\text{cube}} + m)g \quad \dots(ii)$$

Subtracting equation (i) from equation (ii), we get

$$\Rightarrow mg = (50)^3 \times g (1 - 0.3) = 125 \times 0.7 \times 10^3 g$$

$$\Rightarrow m = 87.5 \text{ kg}$$

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Q7

$$(1) \quad P_1 = P_0 + \rho g d_1$$

$$P_2 = P_0 + \rho g d_2$$

$$\Delta P = P_2 - P_1 = \rho g \Delta d$$

$$3.03 \times 10^6 = 10^3 \times 10 \times \Delta d$$

$$\Rightarrow \Delta d \approx 300 \text{ m}$$

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Q8

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(2) Using Bernoulli's equation

$$P + \frac{1}{2}(\rho v_1^2 - \rho v_2^2) + \rho gh = P$$

$$\Rightarrow v_2^2 = v_1^2 + 2gh$$

$$\Rightarrow v_2 = \sqrt{v_1^2 + 2gh}$$

Equation of continuity

$$A_1 v_1 = A_2 v_2$$

$$(1 \text{ cm}^2) (1 \text{ m/s}) = (A_2) \left(\sqrt{(1)^2 + 2 \times 10 \times \frac{15}{100}} \right)$$

$$10^{-4} \times 1 = A_2 \times 2$$

$$\therefore A_2 = \frac{10^{-4}}{2} = 5 \times 10^{-5} \text{ m}^2$$

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Q9

$$(1) \quad 27 \times \frac{4}{3} \pi r^3 = \frac{4}{3} \pi R^3$$

$$\text{or } r = \frac{R}{3}$$

Terminal velocity, $v \propto r^2$

$$\therefore \frac{v_1}{v_2} = \frac{r_1^2}{r_2^2}$$

$$\text{or } v_2 = \left(\frac{r_2}{r_1}\right)^2 v_1 = \left(\frac{R/3}{R}\right)^2 v_1 = \frac{1}{9} v_1$$

$$\text{or } \frac{v_1}{v_2} = 9.$$

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